

EXTRACTING METALS

Starter/Introduction: Metals are very rarely found in a usable form. They normally occur in an ore, that is, as part of a piece of rock. How do you think, then, that man first discovered metals?

ANSWER: *Since nobody knows for certain how man discovered metals, there is obviously no one right answer here. One such explanation is that rocks containing ores were used by Early Man to build fires on. In some cases the heat of the fire may have caused smelting which means separation of the metal from its ore. Presumably Early Man noticed this, and began experimenting with heating up these and other kinds of rocks, until eventually the metals we know today were discovered.*

TEXT:

Separating a metal from other minerals in the ore is known as extraction or smelting. Most metals are smelted using heat, although some, e.g. aluminium are extracted by an electrical process.

Iron is smelted in a tall metal tower, called a blast furnace. The tower is lined with fire- brick¹ and is normally kept burning continuously for several years. Four ingredients are needed: iron ore, coke, limestone and hot air.

A mixture of crushed iron ore, coke and limestone is taken in a skip² up a ramp and fed into the top of the furnace. Hot air is blasted into the base of the fire to produce a very high temperature (1,800°C*).

The smelting process produces three substances: gas, molten ore and slag. The gases escape through an outlet at the top of the furnace. The liquid iron settles at the bottom of the tower. The slag, which consists of the molten limestone and all the impurities it, has absorbed, also runs down to the bottom; but, since it is lighter than the liquid iron, it floats on top of it. Periodically, the iron and the slag are drained off through valves at the bottom of the tower.

When the iron leaves the furnace, it still contains some impurities, particularly carbon. Some of the molten iron is run off into large molds* called pigs, where it is cooled ready for further refining and processing into cast iron at a later stage. The

remainder is taken away in its molten state for further processing into wrought iron³ or steel.

Stages in The Description Of Iron Smelting:

There are three stages in the description of iron smelting:

1. General principle = 1st paragraph.
2. Equipment and Ingredients= 2nd paragraph.
3. .Process= 3rd and 4th paragraph.

There are three stages in description of the process too:

- ***Filling***= 3rd paragraph, from " A mixture of crushed ... top of the furnace"
- ***Blasting***= 3rd – 4th paragraph, from "Hot air is blasted ... floats on top of it."
- ***Emptying***= the rest of the text

Footnotes:

Fire- brick = a kind of brick used to cover the inside of a furnace=
briques réfractaires (Fr)

A skip = a container which is open at the top = **benne** (Fr)

Wrought iron = iron which has been beaten into shape = **fer forgé**.